

Abu Bakar

Full Stack AI Engineer



+923480472778

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✉ abubakarp789@gmail.com

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🌐 LinkedIn

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🐙 GitHub

SUMMARY

Computer Science student and passionate Machine Learning Developer with expertise in machine learning and cloud computing. Experienced in developing end-to-end ML solutions from data preprocessing to deployment using Python, TensorFlow, and cloud technologies. Proven track record through multiple production-ready projects solving real-world problems.

EDUCATION

Namal University Mianwali

Bachelor of Science in Computer Science

FYP: SudhaarAI - Generative Code Assistant and Context-Based Code Optimizer

• Relevant Courses: Artificial Intelligence, Machine Learning, Natural Language Processing, Web Application Development

Oct. 2022 – July, 2026 (expected)

Mianwali, Pakistan

EXPERIENCE

Machine Learning Fellow

Bytewise Limited

• Completed 100-day intensive program covering Python fundamentals, DSA, and complete ML pipeline from data preprocessing to model deployment using Pandas, NumPy, and Scikit-learn

• Built neural networks from scratch using NumPy for MNIST classification and implemented RNNs with TensorFlow for sentiment analysis, mastering backpropagation and sequence modeling

• Developed and deployed 5 production-ready ML applications including F1 Race Predictor, Customer Churn system using Flask/Streamlit with real-time prediction capabilities

June. 2024 – Sep. 2024

Remote

PROJECTS

SudhaarAI | Python, Flask, Next.js, RAG, ChromaDB, MongoDB

• Architected an AI-powered code optimization platform utilizing a decoupled backend Flask REST API and Next.js 16 frontend.

• Engineered a sophisticated 3-phase RAG (Retrieval-Augmented Generation) pipeline orchestrating content extraction, ChromaDB vector database embeddings, query routing, and contextual reranking.

• Designed a responsive UI incorporating Tailwind CSS 4, Radix primitives, and Framer Motion, utilizing Zustand for state management and Clerk for secure JWT-based authentication.

FYP, Namal University

Cotton Seed Classifier | Python, Deep Learning, Flask, Computer Vision

• Built full-stack web application using CNN (ResNet50) model to classify 7 cotton seed varieties with high accuracy

• Developed Flask-based interface with image upload, batch processing, and real-time prediction visualization using Chart.js

• Created cross-platform Flutter mobile application for cotton seed analysis, extending accessibility for field researchers and farmers

Nisar Aziz Agritech Centre, Namal

Customer Churn Predictor | Python, Flask, Machine Learning, Bootstrap

• Developed end-to-end Flask web application predicting customer churn using Random Forest with interactive dashboard

• Built comprehensive ML pipeline including feature engineering, model training, and real-time prediction API

• Created responsive Bootstrap 5 interface with data visualization dashboard showing feature importance and correlation analysis

Bytewise Fellowship

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML and CSS

Frameworks & Libraries: Numpy, Pandas, Matplotlib, Seaborn, Plotly, Scikitlearn, Tensorflow, Pytorch and Flask

Database: MySQL, Neo4j, MongoDB, Azure Database

Tools: Git, GitHub, Microsoft Office, Microsoft Azure

CERTIFICATIONS & ACHIEVEMENTS

Certifications:

• AI For Everyone, DeepLearning.AI, Mar. 2025

• Machine Learning Fellowship, Bytewise Limited, Sep. 2024

• AI Fundamentals, DataCamp, Aug. 2024

• Programming For Everybody (Python), University of Michigan, Jan. 2023

Achievements:

• 2nd Runner-Up, Agri Fusion, 2025 - Developed a cotton seed classification project. Achieved 3rd place out of 35 final year projects as 3rd year student.

• Head of Skill Development Wing, NSSI Society Namal

• Beta Microsoft Learn Student Ambassador (2024 - Present)

• Completed Hacktoberfest 2024 with accepted PRs

• Google Cloud AI Study Jam (Season 6), 31 badges completed